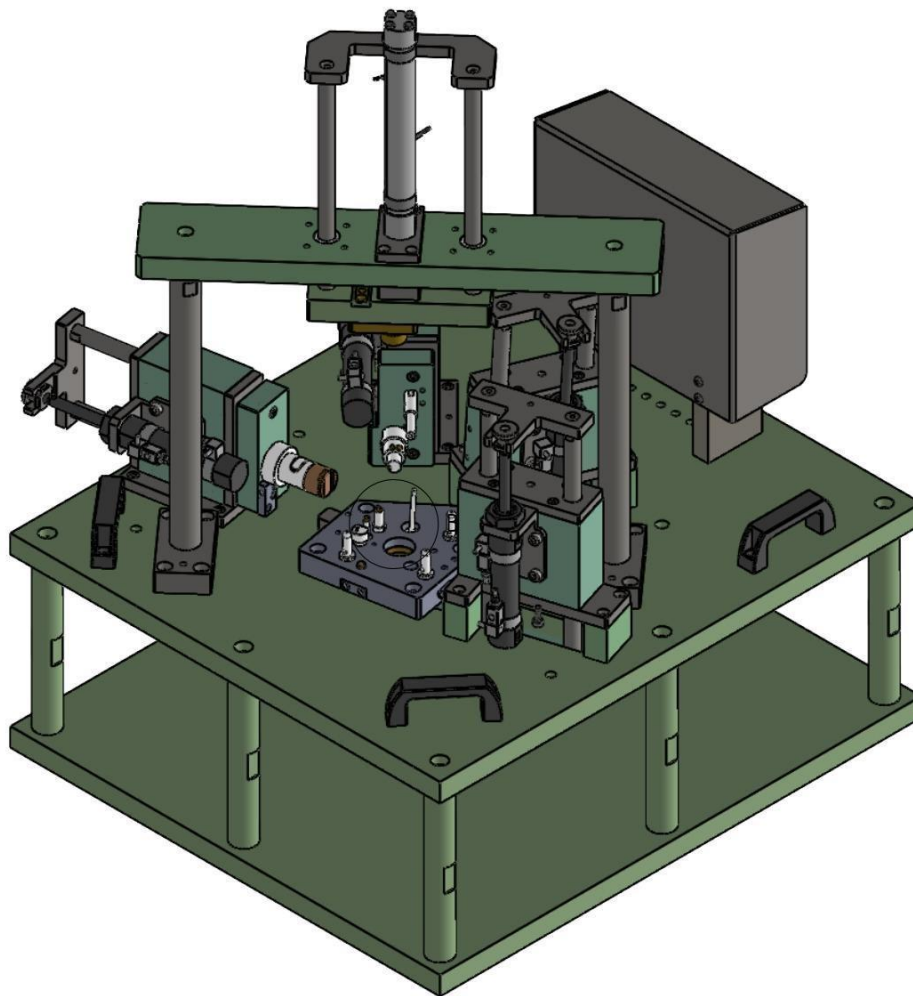


OPERATION AND MAINTENANCE MANUAL

4.1 FIXTURE



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MINDA CORPORATION LTD

DOCUMENT INFORMATION

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1. GENERAL INFORMATION'S

- ❖ This manual provides comprehensive guidelines for the operation, maintenance, and troubleshooting of ECM (Electro chemical Machining) fixtures.
- ❖ It is essential for operators and maintenance personnel to thoroughly understand and follow the instructions in this manual to ensure safe and efficient operation of the fixtures.

2. GENERAL SAFETY WARNINGS

- ❖ **Personal Protective Equipment (PPE):** Always wear appropriate PPE, including safety goggles, gloves, and protective clothing, when operating ECM fixtures.
- ❖ **Electrical Safety:** Ensure that the power supply is turned off before performing any maintenance. Use insulated tools and follow lockout/tag-out procedures.
- ❖ **Chemical Handling:** Handle electrolytes and other chemicals with care. Follow the Material Safety Data Sheets (MSDS) for proper storage, handling, and disposal.
- ❖ **Emergency Procedures:** Familiarize yourself with emergency shut-off procedures, first aid, and fire safety protocols.

3. OVERVIEW OF ECM FIXTURES

- ❖ ECM fixtures are designed to hold and position workpieces during the electrochemical machining process.
- ❖ They play a critical role in ensuring precision and consistency in machining operations.
- ❖ The fixtures are typically made from non-corrosive materials and are equipped with features to facilitate easy workpiece mounting, alignment, and removal.

4. GENERAL OPERATION GUIDELINES

4.1. Pre-Operation Checklist

- ❖ Inspect the ECM fixture for any visible damage or wear.
- ❖ Ensure that all connections, including electrical and electrolyte supply lines, are secure.
- ❖ Verify that the workpiece is correctly positioned and securely clamped in the fixture.
- ❖ Check that the electrolyte flow system is functioning correctly and that the electrolyte level is sufficient.
- ❖ Ensure that the power supply and control systems are functioning properly.

4.2. Machining Process

4.2.1 Cycle Start-Up

- ❖ Turn on the ECM machine and initialize the fixtures.
- ❖ Select the appropriate machining parameters (Voltage, Current, Electrolyte Flow, Machining Time) as specified for the workpiece.
- ❖ Begin the Machining process by starting the auto cycle.

4.2.2 Machining

- ❖ Monitor the machining process closely, ensuring that the electrolyte flow is consistent and that there are no blockages.
- ❖ Observe the fixture and workpiece for any signs of abnormal wear, vibration, or overheating.
- ❖ Make adjustments to the process parameters if necessary to maintain optimal machining conditions.

4.2.3 Cycle End

- ❖ Upon completion of the machining process, the auto end and wait for next cycle.
- ❖ Carefully remove the workpiece from the fixture, ensuring that no damage occurs during removal.

5. MAINTENANCE PROCEDURES

5.1 Regular Maintenance

5.1.1 Daily

- ❖ Clean the fixture and surrounding area to remove any residual electrolyte or debris.
- ❖ Inspect the fixture for signs of corrosion, wear, or damage. Replace any worn or damaged components immediately.

5.1.2 Weekly

- ❖ Check the alignment and condition of clamping mechanisms. Lubricate moving parts as required.
- ❖ Inspect electrical connections for signs of wear or corrosion. Tighten or replace connectors as needed.

5.1.3 Monthly

- ❖ Perform a thorough inspection of the entire fixture, including all electrical and mechanical components.
- ❖ Test the electrolyte flow system for proper operation. Replace filters and clean lines as necessary.

5.2 Preventive Maintenance

- ❖ Schedule periodic checks of the ECM machine and fixtures by qualified personnel.
- ❖ Replace worn components according to the manufacturer's recommendations.
- ❖ Keep a detailed maintenance log to track the condition of the fixture and any repairs or replacements performed.

6. TROUBLESHOOTING

6.1 Fixture Not Holding Workpiece

- ❖ Check for wear or damage to clamping mechanisms.
- ❖ Ensure the workpiece is properly aligned and seated in the fixture.

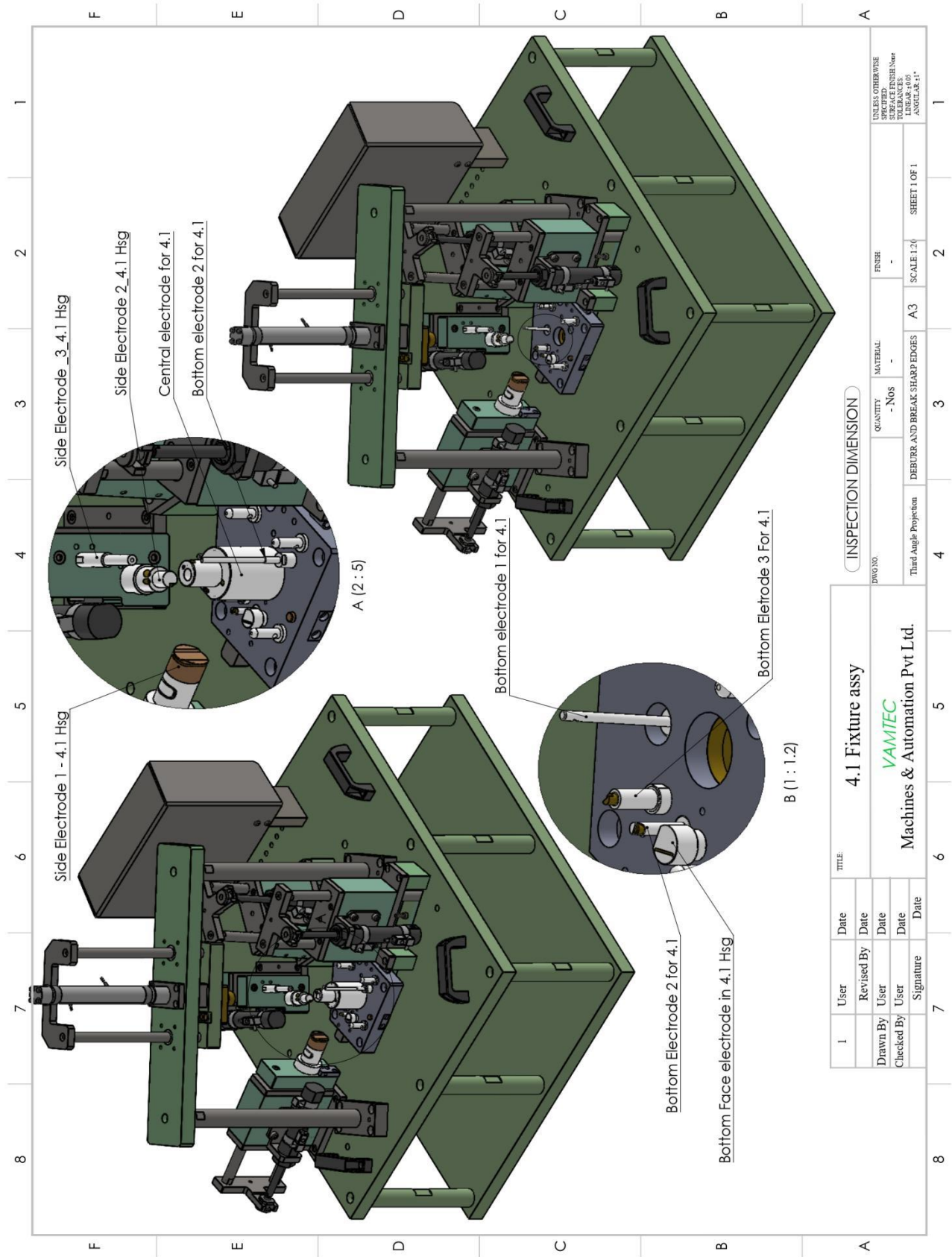
6.2 Electrolyte Flow Issues

- ❖ Inspect for blockages in the electrolyte lines and electrolyte fittings.
- ❖ Check the electrolyte pump operation.

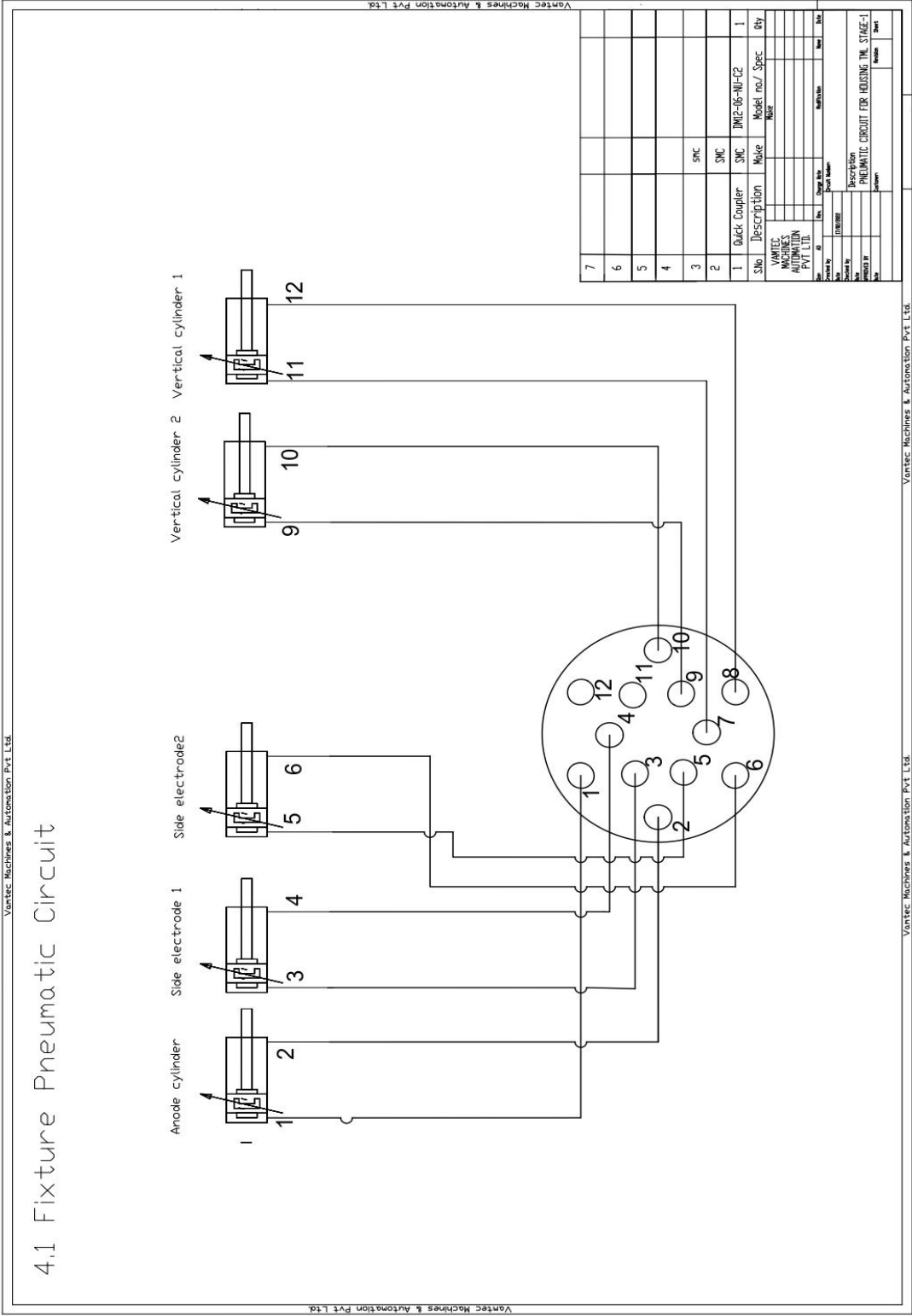
6.3 Short Circuit Issues

- ❖ Check the burr in the workpiece.
- ❖ Check the alignment of the workpiece and electrodes.

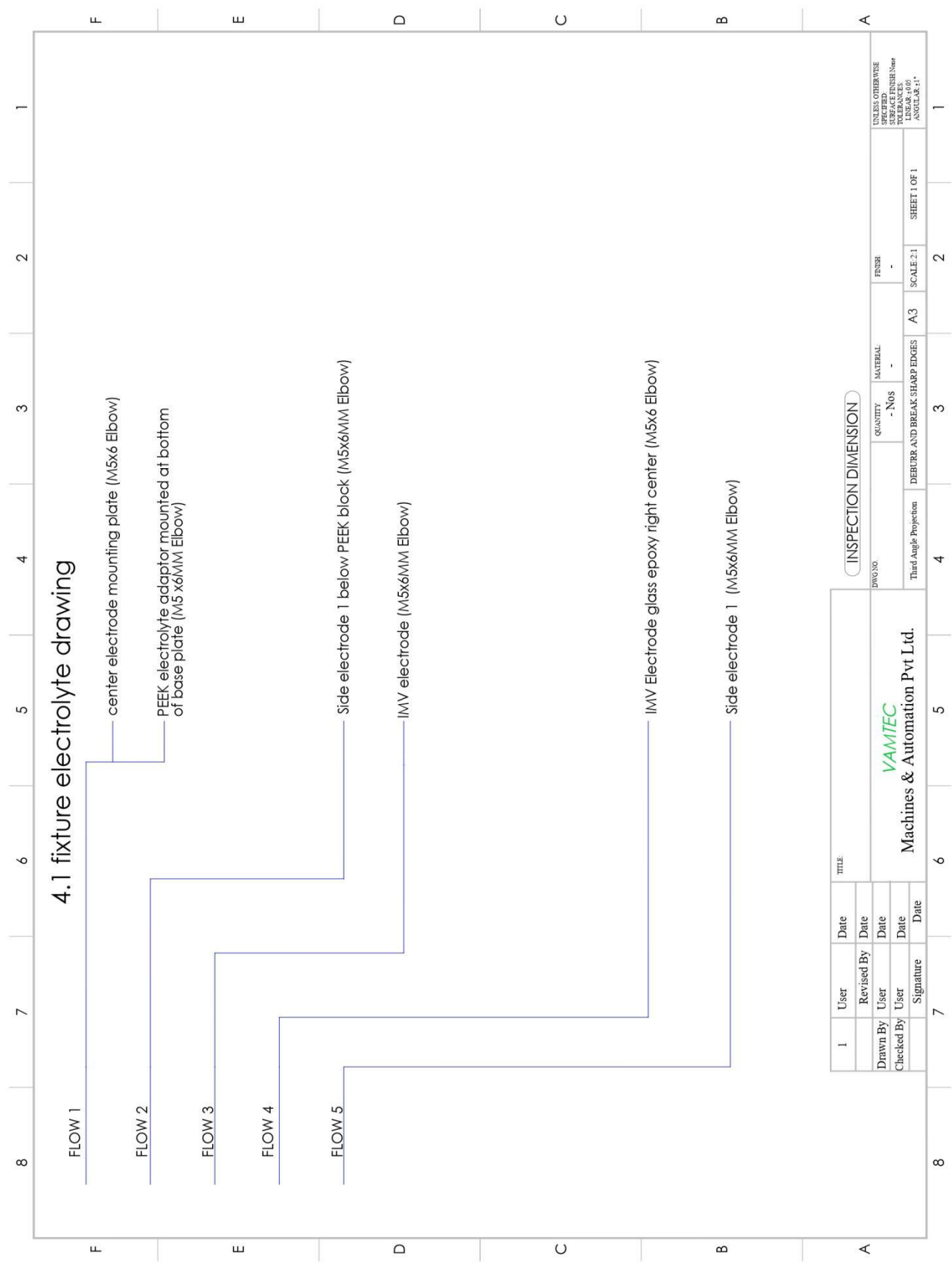
7. FIXTURE DRAWINGS



8. PNEUMATIC CONNECTION DETAILS



9. ELECTROLYTE CONNECTION DETAILS



10. SSR CONNECTION DETAILS

SSR.NO	ELECTRODE NAME
SSR - 1	Bottom Electrode - 3
SSR - 2	Bottom Face Electrode & Bottom Electrode - 2
SSR - 3	-
SSR - 4	Bottom Electrode - 2
SSR - 5	Central Electrode
SSR - 6	-
SSR - 7	Side Electrode - 1
SSR - 8	Side Electrode - 2 & Side Electrode - 3
SSR - 9	Bottom Electrode - 1
SSR - 10	-

11. WEAR PARTS LIST

S.NO	Part Name	Drawing number
1	Peek bush_1	VAM-P-464-01-16 -00
2	Peek Bush for anode guide rod	Vam-P-452-03-05 -00
3	Peek Bush in electrode mounting block	VAM-P-427-03-09-00
4	PEEK Guide For Bottom Electrode	Vam-P-452-03-52 -01
5	Resting Pad	Vam-P-452-03-07 -00
6	Peek Guide Bush For Anode	Vam-P-452-03-51 -00
7	Anode guide bush for 2.2	Vam-P-459-01-13 -00
8	Peek Bush for anode guide rod for 4.1 Hsg	VAM-P-452-05-06-00